Re-analysis Report

Capstone

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1 Assessing Imputation Uncertainty

1.1 Aim of the imputation analysis

In the original publication, the authors applied a single `missForest` imputation on data that were missing no more than 2 below-ground and no more than 3 above-ground traits, thereby increasing the number of species from 301 to 1,218. The goal of the imputation analysis is to assess how sensitive our conclusions about the functional trait space are to the way missing data are handled and to the remaining uncertainty in imputed values.

More specifically, we aim to: (i) test whether the geometry of the functional space (PCA axes, trait loadings, species gradients) is robust to replacing the original single `missForest` imputation with a deliberately naive mean-imputation baseline; (ii) quantify how `missForest` out-of-bag (OOB) imputation uncertainty propagates into species scores and trait loadings in PCA space; (iii) evaluate a conservative worst-case scenario by doubling OOB-based uncertainty, to check whether our main ordination-based conclusions remain stable even under inflated imputation noise.

These aims are implemented in three complementary steps, corresponding to the following R scripts:

- `code/061_multiple_imputation.R` — parametric bootstrap around `missForest` OOB error ("1× OOB")
- `code/062_imputation_mean_simple.R` — naive mean imputation baseline
- `code/063_multiple_imputation_2x_OOB.R` — sensitivity analysis with doubled OOB uncertainty ("2× OOB")

1.2 OOB-Error

`missForest` (Stekhoven, 2011) is a random forest–based imputation algorithm. Like standard random forests, it uses bootstrap samples of the data to grow each tree: for every tree, a bootstrap sample of observations is drawn (with replacement) to build the tree, and the remaining observations are "out-of-bag" (OOB) for that tree. For an observation, an OOB prediction is obtained by aggregating predictions from all trees for which this observation was not used in tree construction. Comparing these OOB predictions to the observed values yields an internal cross-validation estimate of prediction error.

In `missForest`, this internal error is reported as an overall OOB error and, if ‘variablewise = TRUE‘, as a separate error for each imputed variable. For continuous variables the error metric is the normalized root mean squared error (NRMSE), i.e. the root mean squared difference between OOB predictions and observed values, divided by the sample standard deviation of the observed values. For categorical variables `missForest` reports the proportion of falsely classified entries (PFC). In our case all imputed traits are continuous, so we use the trait-wise NRMSEs returned by `missForest` as measures of imputation uncertainty for each trait on the log10 scale.

In the OOB-bootstrap part of our analysis (Step 2 and 3), we convert these trait-wise NRMSEs back to absolute error scales by multiplying them with the observed trait standard deviation. This gives one noise standard deviation per trait (an estimated RMSE), which we then use to generate parametric noise around the single imputed values: for each originally missing entry of trait, we draw bootstrap values from a normal distribution centered on the `missForest` imputation with standard deviation equal to this trait-specific error. This procedure allows us to propagate the uncertainty quantified by the `missForest` OOB error through the PCA and all downstream summaries.

1.3 Step 1: Naive mean imputation

Why do we do this step? We performed a naive mean imputation to assess whether the global PCA structure is robust to the choice of imputation method. The mean imputation ignores phylogeny and multivariate structure and simply replaces each missing value by the trait-wise mean. If PCA results under mean imputation are very similar to those under missForest, this suggests that the ordination is mainly driven by the observed data structure and is not overly sensitive to sophisticated imputation. If results differ strongly, this indicates that missForest (and the information it uses, including phylogeny) meaningfully changes the functional space and therefore needs to be justified more carefully.

How do we do it? In 'code/062_imputation_mean_simple.R', we take the trait matrix and selected traits used in the original PCA, replace all missing entries by their trait-wise means computed from the observed values, and then re-run exactly the same PCA pipeline as in the main analysis (component selection with 'paran', varimax-rotated 'psych::principal', orientation rules, and saving the resulting PCA object). The core steps are sketched below:

```r
library(psych)
library(ks)
library(paran)
traitsUse         <- readRDS("data/traitsUse.rds")
traitsSelect      <- readRDS("data/traitsSelect.rds")
AllTraitsAllInfo <- readRDS("data/imputedTraits.rds")

## Mean imputation
trait_means <- sapply(traitsUse[, traitsSelect], function(col) mean(col
    , na.rm = TRUE))
imputedTraits_mean <- traitsUse[, traitsSelect]
for (j in colnames(imputedTraits_mean)) {
  imputedTraits_mean[is.na(imputedTraits_mean[, j]), j] <- trait_means[
      j]
}
AllTraitsAllInfo_mean <- AllTraitsAllInfo
AllTraitsAllInfo_mean[, traitsSelect] <- imputedTraits_mean
saveRDS(AllTraitsAllInfo_mean, "data/imputedTraits_mean.rds")

## PCA on mean-imputed traits (same pipeline as main analysis)
AllTraits_mean             <- AllTraitsAllInfo_mean[, traitsSelect]
PCATotal_mean              <- list()
PCATotal_mean$traits       <- AllTraits_mean
PCATotal_mean$dimensions   <- paran(PCATotal_mean$traits)$Retained
PCATotal_mean$PCA <- psych::principal(scale(PCATotal_mean$traits),
                                      nfactors = PCATotal_mean$
                                          dimensions,
                                      rotate   = "varimax", covar =
                                          TRUE)

# Orientation rules and TPD calculations follow the main script
saveRDS(PCATotal_mean, "data/PCATotal_mean_imputation.rds")
```

Listing 1: Core analysis steps to perform naive mean imputation (schematic).

What do we learn from this step? The major axes of variation and trait loadings under mean imputation are very similar to those under missForest. Species positions in the first four PCA dimensions show only small shifts; global gradients and group separations are preserved (Fig. 1). This indicates that the geometry of the functional trait space is robust to replacing missForest with a much simpler imputation, and that the main ordination-based conclusions do not depend critically on the specific imputation algorithm.

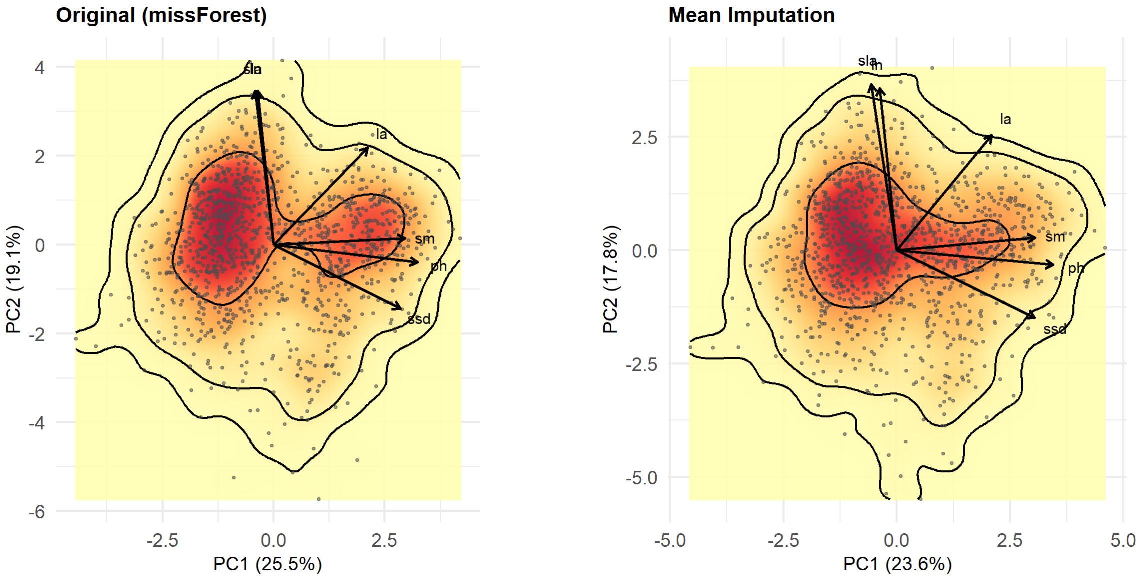

Figure 1: Functional space (PC1–PC2) for original data (left) and mean-imputed data (right).

1.4 Step 2: Propagating OOB imputation uncertainty

Why do we do this step? A single `missForest` imputation gives one "best guess" dataset and treats imputed values as fixed, so downstream analyses ignore imputation variance and may understate uncertainty. `missForest` provides trait-wise OOB error estimates that quantify the typical prediction error for each trait (see OOB section above). In this step we turn these error estimates into a parametric bootstrap around the single imputed dataset. By repeatedly perturbing only the originally missing entries according to their trait-specific OOB error and re-running the full PCA pipeline, we obtain a direct measure of how much imputation uncertainty propagates into species scores and trait loadings.

How do we do it? In ‘code/061_multiple_imputation.R‘, we first run `missForest` with ‘variablewise = TRUE‘ to obtain a single imputed dataset together with trait-wise OOB NRMSE values. We then convert these normalized errors back to absolute error scales, identify which entries were originally missing, and, for each of 50 bootstrap replicates, add Gaussian noise with trait-specific SD only to those missing cells. For every perturbed dataset we re-run the full PCA/TPD pipeline using the same helper as in the main analysis, and finally summarize the resulting lists of scores and loadings across bootstraps (means and SDs) and save them for downstream use. The key implementation steps are:

```
library(missForest)

## Single missForest imputation + OOB errors
traitsAux          <- readRDS("data/traitsAux.rds")
traitsSelect       <- readRDS("data/traitsSelect.rds")
AllTraitsAllInfo <- readRDS("data/imputedTraits.rds")

mf     <- missForest(xmis = traitsAux, variablewise = TRUE, verbose =
    TRUE)
ximp0 <- mf$ximp
saveRDS(mf, "data/imputedTraitsOOB.rds")
```

```r

oob_norm <- mf$OOBerror[traitsSelect]                # trait-wise NRMSE
sd_obs   <- sapply(traitsSelect, function(j) sd(traitsAux[, j], na.rm = TRUE))
noise_sd <- oob_norm * sd_obs                        # RMSE per trait
noise_sd[is.na(noise_sd) | noise_sd <= 0] <- 1e-8
missing_mat <- is.na(as.matrix(traitsAux))           # originally missing entries

## Parametric OOB bootstrap around the single imputation
nboot  <- 50
outdir <- "data/imputed_bootstrap"
dir.create(outdir, showWarnings = FALSE, recursive = TRUE)

scores_list   <- vector("list", nboot + 1)
loadings_list <- vector("list", nboot + 1)

AllTraitsAllInfo_single <- AllTraitsAllInfo
AllTraitsAllInfo_single[, traitsSelect] <- ximp0[, traitsSelect]
saveRDS(AllTraitsAllInfo_single, file.path(outdir, "imputed_single.rds"))

PCA_single          <- make_PCAtotal_from_AllTraits(AllTraitsAllInfo_single, suffix = "_single")
scores_list[[1]]    <- PCA_single$PCA$scores
loadings_list[[1]]  <- PCA_single$PCA$loadings[, 1:4]

for (b in seq_len(nboot)) {
  set.seed(10000 + b)
  xboot <- ximp0
  for (j in seq_len(ncol(xboot))) {
    miss_idx <- which(missing_mat[, j])
    if (length(miss_idx) > 0) {
      xboot[miss_idx, j] <- rnorm(length(miss_idx),
                                  mean = ximp0[miss_idx, j],
                                  sd   = noise_sd[j])
    }
  }
  AllTraitsAllInfo_b <- AllTraitsAllInfo
  AllTraitsAllInfo_b[, traitsSelect] <- xboot[, traitsSelect]
  saveRDS(AllTraitsAllInfo_b, file.path(outdir, sprintf("imputed_boot_%03d.rds", b)))

  PCAb                 <- make_PCAtotal_from_AllTraits(AllTraitsAllInfo_b, suffix = sprintf("_boot_%03d", b))
  scores_list[[b + 1]] <- PCAb$PCA$scores
  loadings_list[[b+1]] <- PCAb$PCA$loadings[, 1:4]
}

## Summaries across bootstraps (not shown in detail here)
saveRDS(list(scores = scores_list, loadings = loadings_list),
        file.path(outdir, "PCA_scores_loadings_boot_raw.rds"))
```

Listing 2: Core analysis steps to perform parametric bootstrap imputation (schematic).

What do we learn from this step? For all four main components, the bootstrap spread of species scores is small relative to the overall range of the ordination axes. `missForest` Mean bootstrap scores are extremely close to the deterministic single-imputation scores, indicating that missForest does not systematically bias species positions (Fig. 2). Trait loadings show very small bootstrap SDs; the identity and interpretation of the main PCA axes are highly stable under imputation uncertainty. Overall, this analysis shows that realistic levels of imputation uncertainty (as quantified by the OOB error) do not alter global gradients, trait axes, or group-level patterns.

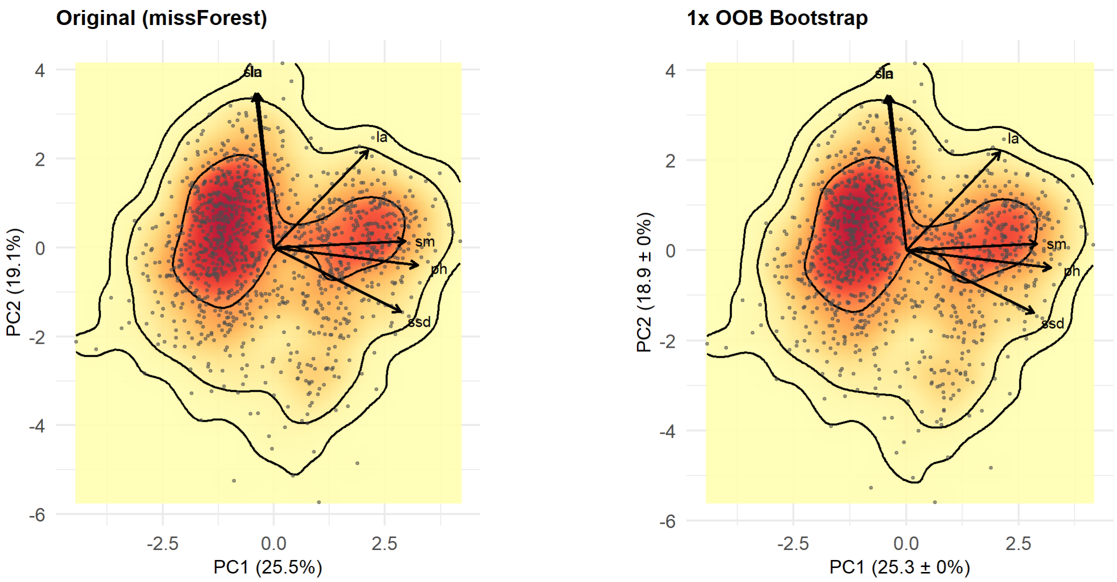

Figure 2: Functional space (PC1–PC2) for the original data (left) and the bootstrap-imputed data (right).

1.5 Step 3: Sensitivity analysis with inflated uncertainty

Why do we do this step? The OOB-based error estimates used in Step 2 are model-based and might, in principle, underestimate the true uncertainty in imputed values. To probe this, we repeat exactly the same OOB-bootstrap procedure but inflate all trait-wise OOB error SDs by a factor of two before generating the perturbations. If PCA scores, loadings and group-level patterns remain essentially unchanged under this amplified noise, we gain evidence that our main ordination conclusions are robust even under a conservative "worst-case" imputation uncertainty.

How do we do it? In ‘code/063_multiple_imputation_2x_OOB.R‘, we repeat the same workflow as in Step 2 but inflate the trait-wise OOB NRMSEs by a factor of two before converting them to absolute error SDs. The subsequent parametric bootstrap, PCA re-fitting and summarisation are identical to the 1× OOB case. The essential code changes are:

```
library(missForest)

traitsAux        <- readRDS("data/traitsAux.rds")
traitsSelect     <- readRDS("data/traitsSelect.rds")
AllTraitsAllInfo <- readRDS("data/imputedTraits.rds")

mf     <- missForest(xmis = traitsAux, variablewise = TRUE, verbose =
    TRUE)
ximp0 <- mf$ximp

## Inflate trait-wise OOB errors by factor 2
oob_raw  <- mf$OOBerror * 2
oob_norm <- oob_raw[traitsSelect]
```

```
sd_obs   <- sapply(traitsSelect, function(j) {
  v <- traitsAux[, j][!is.na(traitsAux[, j])]
  if (length(v) > 1) sd(v) else 0
})

noise_sd <- oob_norm * sd_obs   # 2x amplified RMSE per trait
noise_sd[is.na(noise_sd) | noise_sd <= 0] <- 1e-8
missing_mat <- is.na(as.matrix(traitsAux))

cat("\n*** OOB-BASED NOISE MULTIPLIED BY 2.0 ***\n")
print(noise_sd[traitsSelect])

## The subsequent bootstrap loop, PCA re-fitting and summaries
## follow exactly the same structure as in code/061_multiple_imputation.R,
## but using these inflated noise_sd values.
```

Listing 3: Core analysis steps to perform parametric bootstrap imputation with inflated OOB scores (schematic).

What do we learn from this step? Doubling the OOB-based noise modestly increases the SD of species scores, but the increase is small relative to the total spread of the PCA axes. Mean species positions under 2× OOB remain extremely close to those under 1× OOB. Trait loadings under 2× OOB remain highly similar to those under 1× OOB, and the qualitative interpretation of PCA axes is unchanged (Fig. 3). The exact ordinations obtained in this impuation analysis will be compared formally in a later step using a Procrustes test.

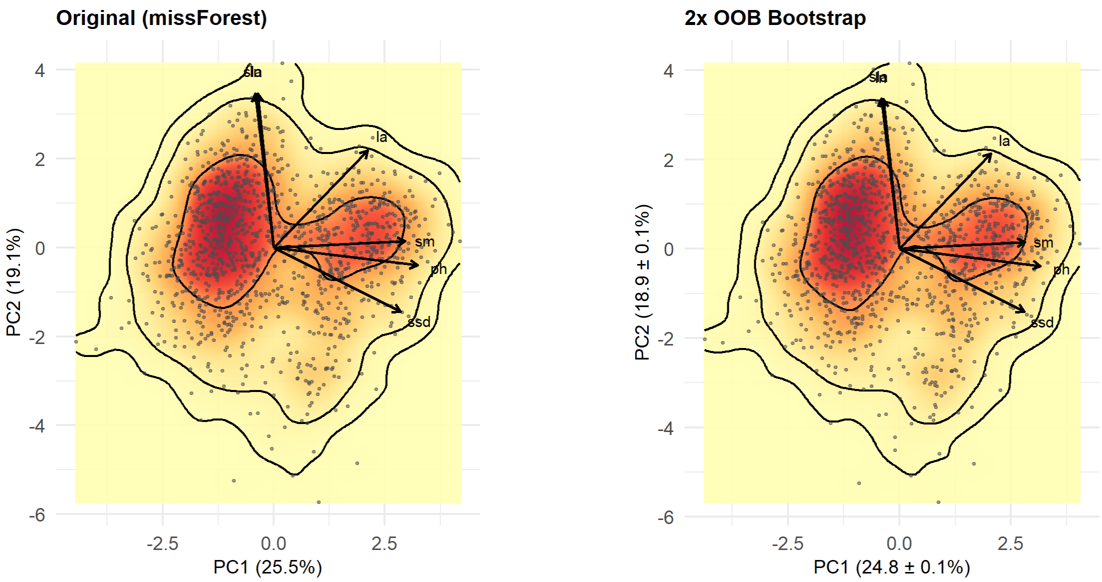

Figure 3: Functional space (PC1–PC2) for the original data (left) and the bootstrap-imputed data with inflated OOB score (right).

1.6 Overall Conclusion

Across the three steps, the re-analysis shows that The global PCA structure is insensitive to the choice between naive mean imputation and missForest. When imputation uncertainty is explicitly propagated using missForest OOB errors, species scores and trait loadings show only modest variation around the deterministic single-imputation solution. Even when OOB-based uncertainty is doubled, the visual ordination geometry and trait axes appear highly stable. Overall, these results strongly suggest that the main findings about the fine-root functional trait space are robust to missing-data treatment and that imputation uncertainty primarily adds small, local noise rather than changing the overall structure of the trait space. A definitive quantitative assessment of how similar the ordinations are will, however, come from the Procrustes comparisons presented in the subsequent analysis.

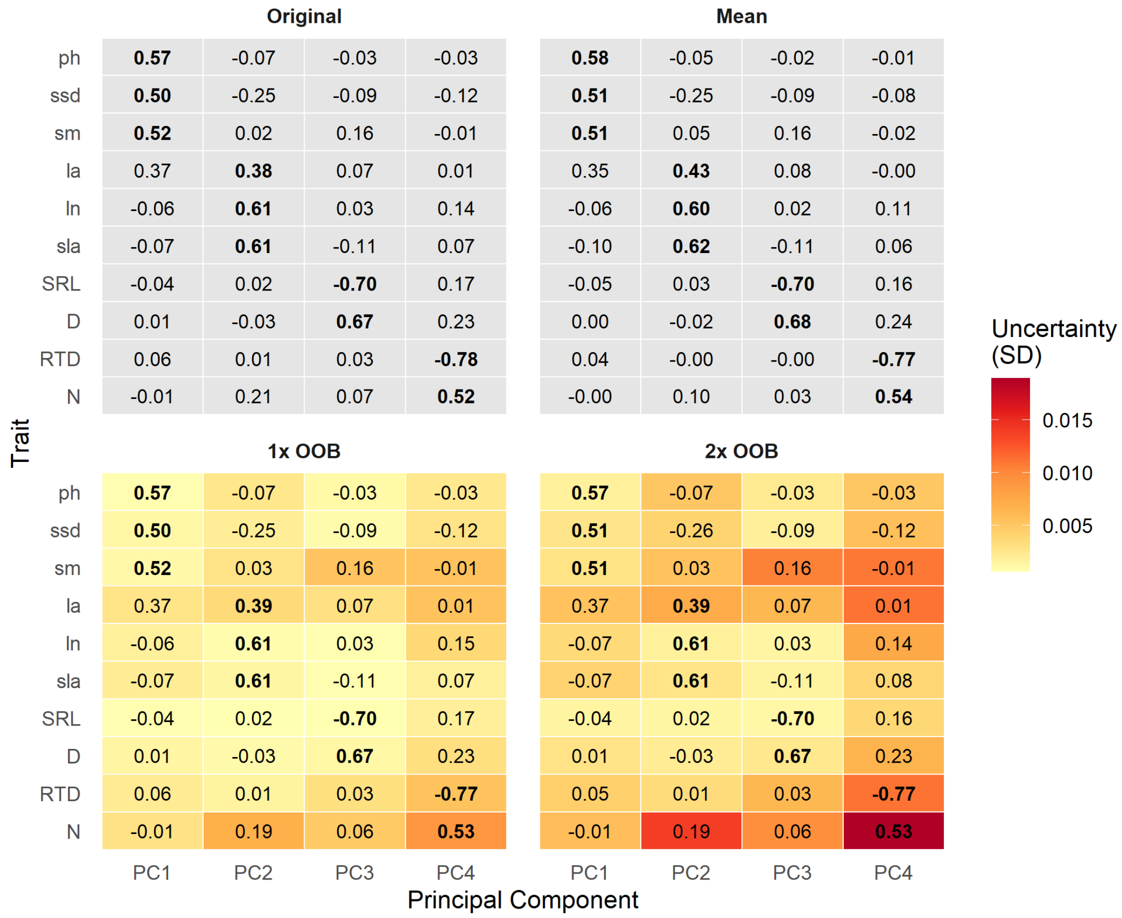

Figure 4: Loadings of functional traits on the first four principal components under four imputation methods. Rows show traits, columns show PCs. Tile color encodes uncertainty (standard deviation of loadings across bootstraps) where available.

2 Assessing Regionality of the Results

2.1 Aim of the biome analysis

The data used in the publication originates from the Try database (Kattge, 2020) and GRooT database (Guerrero-Ramirez, 2021). Carmona et al. (2021) found that 76% of the variance of this combined data can be explained by four principle components of a PCA. The above traits are thereby located on the first two components, the aboveground plane, while the below traits are related to two components, the belowground plane. On these four dimensions three trait hotspots were identified, visualizing two plant strategies aboveground, associated with herbaceous species or angiosperm trees, and one belowground trait space. Supporting their research with PERMANOVA tests to analyse the dissimilarity between species groups, Carmona et al. (2021) reason that plant species partition the above ground trait space, while sharing the fine-root trait space. The PERMANOVA analysis includes a biome comparison, highlighting dissimilarities of 42.8% ± 18.1 aboveground and of 21.6% ± 6.8 belowground.
The aim of this biome specific analysis is to verify the global university of the publication results Carmona et al. (2021), by implementing a biome-specific PCA analysis. Instead of relying on dissimilarity results from PERMANOVA, the higher variation on the aboveground plane will be visually identified via biome-specific PCAs for the three biomes, temperate, continental and tropical.

The biome analysis is implemented in three steps, corresponding to the following R scripts:

- `Code/PCA_preparation_biome_subdivision.R`
- `Code/PCA_reanalysis.R`
- `Code/PCA_visualization.R`

2.2 Step Data separation by biome

For this analysis the data needs to be sorted into data subsets, each containing only species of one biome. The root trait data (GRooT) includes the global biome information, revealing a distinction in the three biomes temperate, continental and tropical. Fig. 5 shows the distribution of species associated with the three biomes, according to the GRooT database.

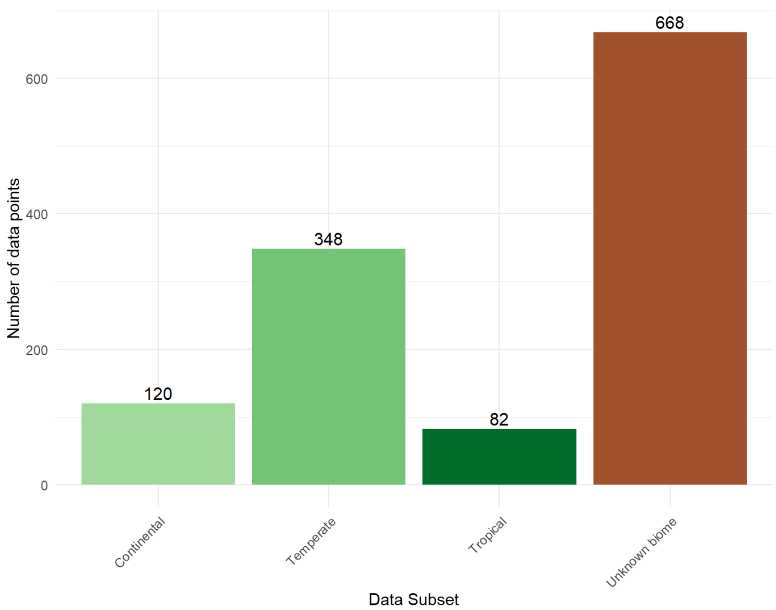

Figure 5: Data amount per biome GRooT division

The low data availability for continental and tropical biome, with 120 and 82 species respectively, lead to the question, if a PCA with such little data is interpretable and comparable. This question will be answered with a PCA stability analysis in Fig. 2.4. Carmona et al. (2021) used a different biome separation in their PERMANOVA tests, which separates the species into nine biomes according to Fig. 6. This biome separation counts species multiple times, as they can occur at multiple locations if they have a broad ecological niche. However, some biomes are drastically under-represented, such as Tundra with only 14 data points. Sorting the second biome separation into the same categories as above leads to a similar biome distribution as observed by the GRooT biome sorting (6 B). However, this categorized version of all nine biomes is not usable do to the extreme lack of data for the continental biome. In the following analysis, we will focus on the GRooT biome distribution and compare this outcome with the separation into the nine biomes shown in Fig. 6 A.

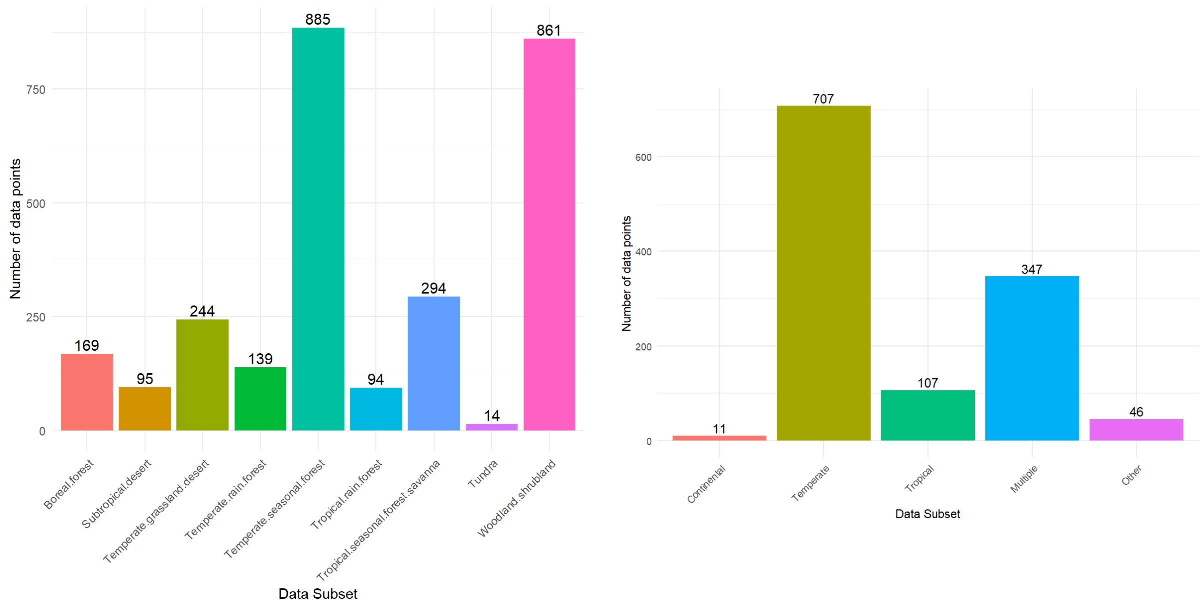

Figure 6: Data amount per biome with A) all nine biomes and B) clustering species into temperate, tropical, continental, multiple and other biomes

2.3 Step Verification of PCA recreation results

Based on the original publication script "06_Imputed information spectrum.R", an R-function was written that conducts a PCA in the same manner as in the publication, allowing to quickly create PCA results for new data subsets. This recreation of PCA uses the same grid size and varimax rotation, as done in the publication. Additionally, this analysis focuses on the use of imputed data, in order to increase the amount of available data. The accuracy of imputation and varimax rotation are further explored in Section 1 and 3.3 and will therefore be taken for granted in this analysis. The PCA function can be seen below, with the exception of axis orientation flipping- and pairwise trait probability density calculation excerpts, which have been removed to show a simplified overview. For the complete function and its application please check the script "PCA_reanalysis.R".

```
# PCA function without axis orientation flipping and pairwise tpd
perform_PCA <- function(traits_df, pca_name,
                        gridSize_full = 30,
                        gridSize_2D = 200) {
  tryCatch({
    cat("\n=== Computing", pca_name, "===\n")
```

```r
    flush.console()
    pca_list <- list()
    pca_list$traits <- traits_df
    pca_list$dimensions <- paran(pca_list$traits)$Retained
    pca_list$dimensions <- min(paran(pca_list$traits)$Retained, 4)
    pca_list$means <- apply(pca_list$traits, 2, mean)
    pca_list$sds <- apply(pca_list$traits, 2, sd)

    # pca computation
    pca_list$PCA <- psych::principal(scale(pca_list$traits), nfactors=pca_list$dimensions,
                                   rotate="varimax", covar = TRUE)
    pca_list$Variance <- pca_list$PCA$Vaccounted[2,]
    # eigenvalue extraction
    sqrtEigen <- sqrt(colSums(pca_list$PCA$loadings**2))
    # score rescaling
    for(i in 1:pca_list$dimensions){
      pca_list$PCA$scores[, i] <- pca_list$PCA$scores[, i] * sqrtEigen[i]
    }

    # Convert loadings to eigenvectors
    sqrtEigenMat <- matrix(rep(sqrtEigen, nrow(pca_list$PCA$loadings)),
                           byrow = TRUE,
                           nrow = nrow(pca_list$PCA$loadings))
    # convert loadings to eigenvectors
    pca_list$PCA$loadings <- pca_list$PCA$loadings / sqrtEigenMat

    # [...] hidden excerpt axis orientation flipping

    # convert discrete PCA scores into probability density functions in trait space
    pca_list$traitsUse <- data.frame(pca_list$PCA$scores[, 1:pca_list$dimensions])
    sdTraits <- sqrt(diag(Hpi.diag(pca_list$traitsUse)))

    pca_list$TPDs <- TPDsMean(
      species = rownames(pca_list$traitsUse),
      means = pca_list$traitsUse,
      sds = matrix(rep(sdTraits, nrow(pca_list$traitsUse)), byrow=TRUE,
                   ncol=pca_list$dimensions),
      n_divisions = gridSize_full
    )
    # [...] hidden excerpt pairwise TPD using gridSize_2D
    }
    output_file <- paste0("PCA_results/", pca_name, ".rds")
    saveRDS(pca_list, output_file)
    cat("Successfully saved:", output_file, "\n")
    return(pca_list)
  })
}
```

Listing 4: Key code lines (Step 3): PCA recreation.

First, a PCA with the imputed data set of 1218 species was repeated to verify the correctness of this recreated PCA function. As shown in Fig. 8 b and Fig. 7, the calculated loadings of the recreated PCA are identical to the publications loadings of the imputed data set with varimax rotation, with negligible differences at the second decimal place. Hereby, the comparability of future PCAs is verfied.

PCA Loadings – Full

Trait	PC1	PC2	PC3	PC4
ph	**0.57**	-0.07	-0.03	-0.03
ssd	**0.50**	-0.25	-0.09	-0.12
sm	**0.52**	0.03	0.16	-0.01
la	0.37	**0.39**	0.07	0.01
ln	-0.06	**0.61**	0.03	0.15
sla	-0.08	**0.61**	-0.11	0.07
SRL	-0.04	0.02	**-0.70**	0.16
D	0.01	-0.03	**0.67**	0.23
RTD	0.06	0.01	0.03	**-0.77**
N	-0.01	0.19	0.06	**0.54**

Figure 7: Loadings PCA recreation imputed data set full.

a PCA

	Full dataset (1,719 sp)				Complete dataset (301 sp)				Imputed dataset (1,218 sp)			
	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4
eigenvalue	2.70	1.96	1.72	1.14	2.81	1.93	1.70	1.16	2.75	2.00	1.75	1.09
% variance	27.0	19.6	17.2	11.4	28.1	19.3	17.0	11.6	27.5	20.0	17.5	10.9
ph	**0.50**	0.15	-0.24	0.12	**0.50**	0.11	-0.22	0.16	**0.50**	0.15	-0.21	0.12
ssd	**0.52**	-0.08	-0.17	0.14	**0.49**	-0.07	-0.18	0.25	**0.51**	-0.06	-0.17	0.14
sm	**0.44**	0.29	-0.12	0.01	**0.43**	0.29	-0.14	0.00	**0.46**	0.27	-0.05	0.05
la	0.16	**0.45**	-0.26	-0.23	0.15	0.38	**-0.41**	-0.19	0.21	**0.46**	-0.21	-0.15
ln	-0.27	**0.44**	-0.23	-0.19	-0.31	**0.44**	-0.29	-0.03	-0.25	**0.48**	-0.16	-0.22
sla	-0.32	**0.35**	-0.29	-0.26	-0.33	0.31	**-0.35**	-0.20	-0.27	**0.41**	-0.29	-0.26
SRL	-0.18	-0.29	**-0.57**	0.26	-0.15	-0.43	**-0.50**	0.25	-0.17	-0.21	**-0.62**	0.29
D	0.08	0.36	**0.60**	0.03	0.06	0.47	**0.51**	0.11	0.07	0.30	**0.62**	0.10
RTD	0.17	-0.25	-0.04	**-0.69**	0.17	-0.13	-0.03	**-0.72**	0.20	-0.21	-0.03	**-0.75**
N	-0.14	0.31	-0.04	**0.51**	-0.20	0.21	-0.07	**0.50**	-0.15	0.33	0.01	**0.40**

b PCA + varimax rotation

	Full dataset (1,719 sp)				Complete dataset (301 sp)				Imputed dataset (1,218 sp)			
	C1	C2	C3	C4	C1	C2	C3	C4	C1	C2	C3	C4
ph	**0.58**	-0.01	-0.05	0.04	**0.57**	-0.09	0.00	-0.04	**0.57**	-0.06	-0.03	-0.04
ssd	**0.51**	-0.22	-0.11	-0.03	**0.52**	-0.24	-0.11	0.01	**0.50**	-0.25	-0.08	-0.12
sm	**0.52**	0.10	0.12	0.02	**0.51**	0.06	0.18	-0.10	**0.52**	0.02	0.16	-0.01
la	0.34	**0.48**	0.07	-0.07	0.34	**0.44**	0.02	-0.12	0.37	**0.40**	0.07	0.01
ln	-0.05	**0.59**	0.01	0.07	-0.08	**0.59**	0.04	0.23	-0.06	**0.61**	0.03	0.15
sla	-0.11	**0.59**	-0.09	-0.01	-0.14	**0.60**	-0.08	0.06	-0.08	**0.61**	-0.11	0.07
SRL	-0.02	0.00	**-0.70**	0.15	-0.04	0.00	**-0.69**	0.16	-0.04	0.02	**-0.70**	0.17
D	-0.03	-0.04	**0.68**	0.16	0.01	-0.02	**0.68**	0.21	0.01	-0.03	**0.67**	0.23
RTD	0.01	0.07	0.00	**-0.75**	0.03	0.02	-0.01	**-0.72**	0.06	0.01	0.03	**-0.77**
N	0.04	0.08	0.00	**0.61**	-0.04	0.16	0.02	**0.57**	0.00	0.19	0.06	**0.53**

c PCA + varimax rotation after exclusion of pot data

	Complete excluding pot data (207 sp)			
	C1	C2	C3	C4
ph	**0.55**	-0.01	-0.08	0.05
ssd	**0.54**	-0.14	-0.04	0.00
sm	**0.50**	0.07	-0.14	-0.05
la	0.34	**0.47**	0.00	-0.11
ln	-0.05	**0.60**	0.06	0.21
sla	-0.08	**0.61**	0.13	0.07
SRL	-0.11	0.13	**-0.62**	0.12
D	0.04	-0.06	**0.69**	0.13
RTD	0.07	-0.08	0.25	**-0.72**
N	0.00	0.07	0.15	**0.72**

Figure 8: Original PCA loadings Carmona et al. (2021)

2.4 Step Examination of PCA stability dependent on data amount

In order to check PCA stability for lower data amounts, the imputed dataset is randomly sorted into subsets, containing half, a quarter and a tenth of the full data amount. A quarter of the full data amount has 44 data points fewer than the largest biome availability, temperate, which contains 348 data points. A tenth of the data is almost identical to the continental data set, both serving as a suitable comparison.

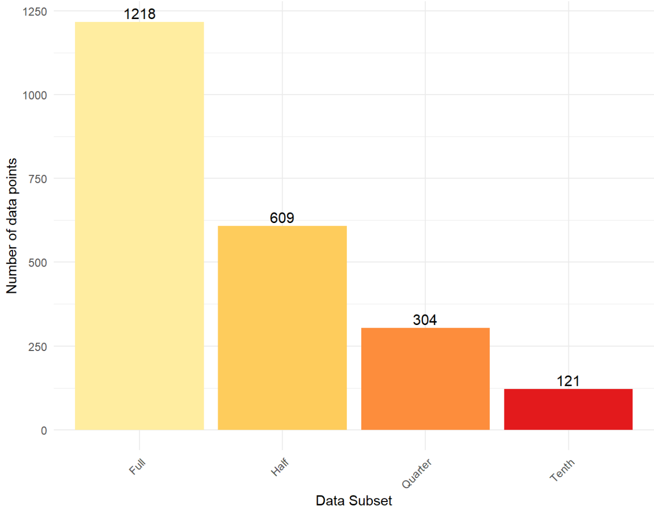

Figure 9: Data amounts of imputed data subsets

The results of this first PCA analysis are shown in Fig. 10 and 11. Unlike publication results (Carmona et al., 2021) all trait loadings are depicted on the PC1-PC2 plane, with root traits in capital letters and above traits in small letters. A and B of Fig. 10, resembling full and half of the data amount respectively, confirm that the aboveground traits are dominant on the so called aboveground plane. The root traits are prevalent on the fine-root plane (PC3-PC4) as shown in Fig. 11. For less data (quarter and half) only two PCA components could be created. Apparently, there was too little data to reliably estimate additional independent directions of variance, so the covariance matrix had a reduced rank and only supported two meaningful principal components instead of four or more. Due to this, the root traits gained importance on the PC1-PC2 plane for a quarter and a tenth of the data, as observed by the length increase of the loadings in Fig. 10 C and D. The loadings listed in AppendixC also shows this shift of root traits to the PC1-PC2 plane. Concluding, data amounts smaller than half of the original 1218 species contain not enough variance to explain more than two principle components. It is therefore not possible to detect a shift of the root traits to the PC3-PC4 plane. As the continental biome contains only 120 and the tropical biome only 82 species, it is not possible to make an unambiguous statement about these two biomes.

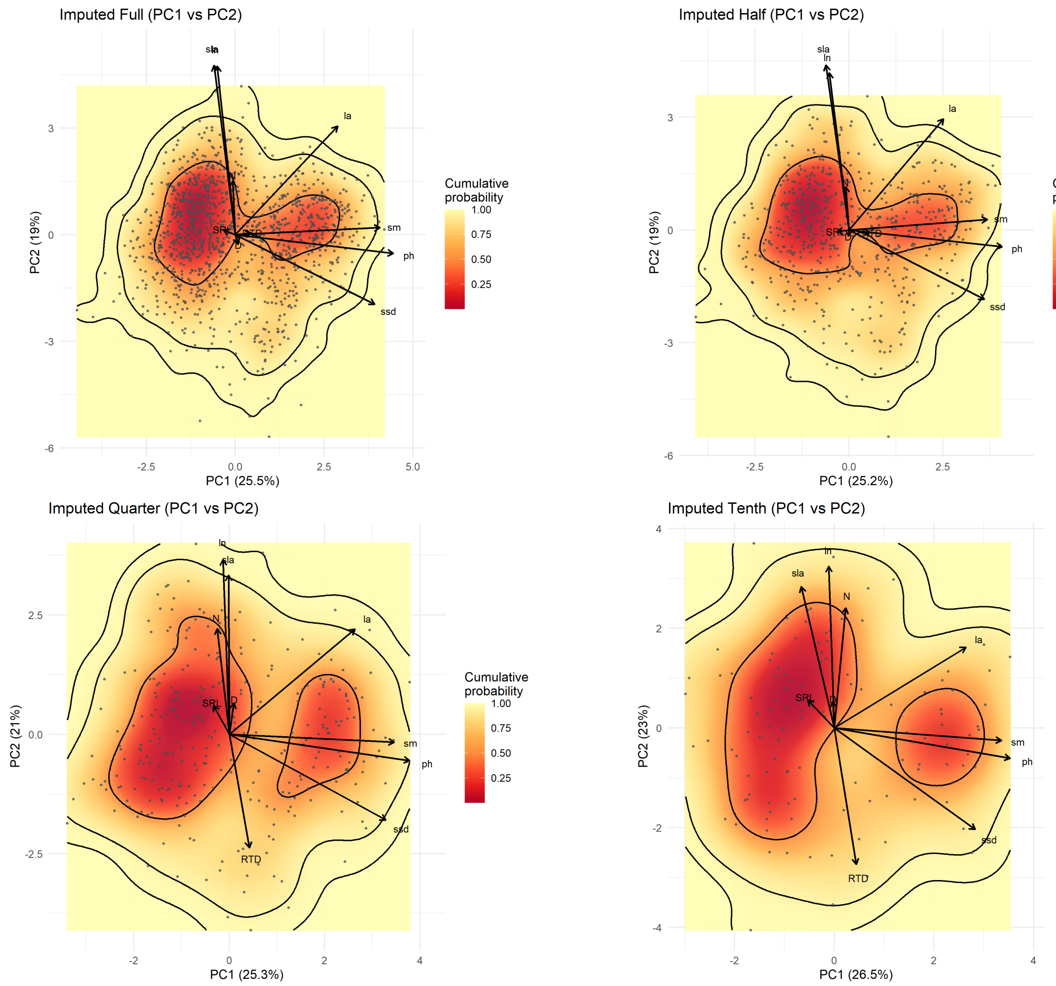

Figure 10: Functional space (PC1–PC2) for full (A), half (B), quarter (C) and tenth (D) of the data amount.

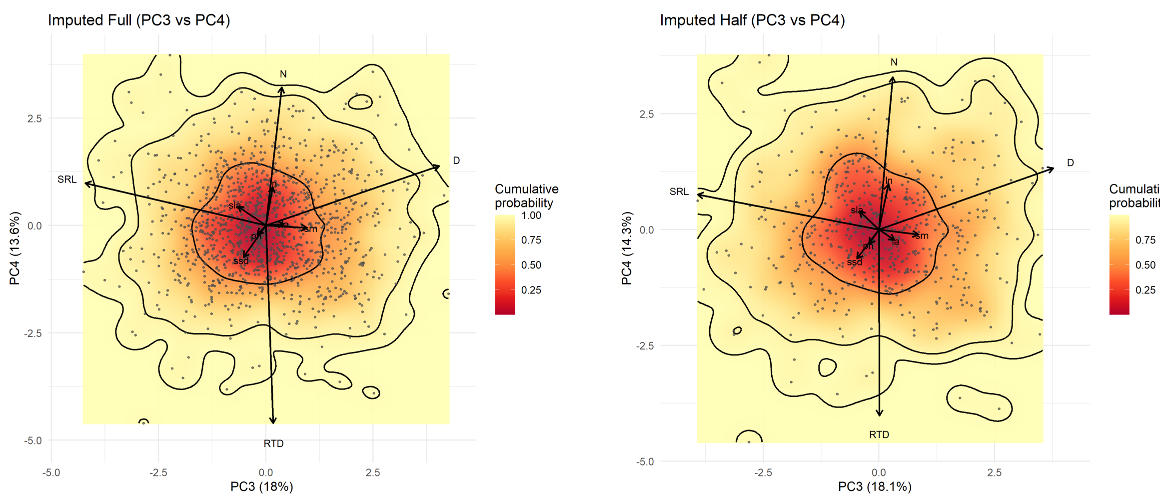

Figure 11: Functional space (PC3–PC4) for full (A) and half (B) of the data amount.

2.5 Step Biome specific PCA analysis

Despite the results of Fig. 2.4, a biome specific PCA analysis was conducted. The results shown in Fig. 12 will now be analyzed with caution. All three biomes, sorted after the GRooT structure, contain equal to or less than 348 species and therefore only resulted in two principle components. A classification of the plant traits to the above- and fine-root plane are therefore not possible. Nevertheless, it is possible to focus on the second main finding of the publication (Carmona et al., 2021), which states that there are more dissimilarities on the aboveground plane, compared to the fine-root plane. The biomes temperate (A) and continental (B) depict two functional spaces, that visualize this variation. If one assumes, that with more data, the root traits would shift to the below ground plane, as observed reversely in the above PCA stability test (2.4), the findings of Carmona et al. (2021) are supported. The third tropical biome (C), however, only depicts one functional trait space.

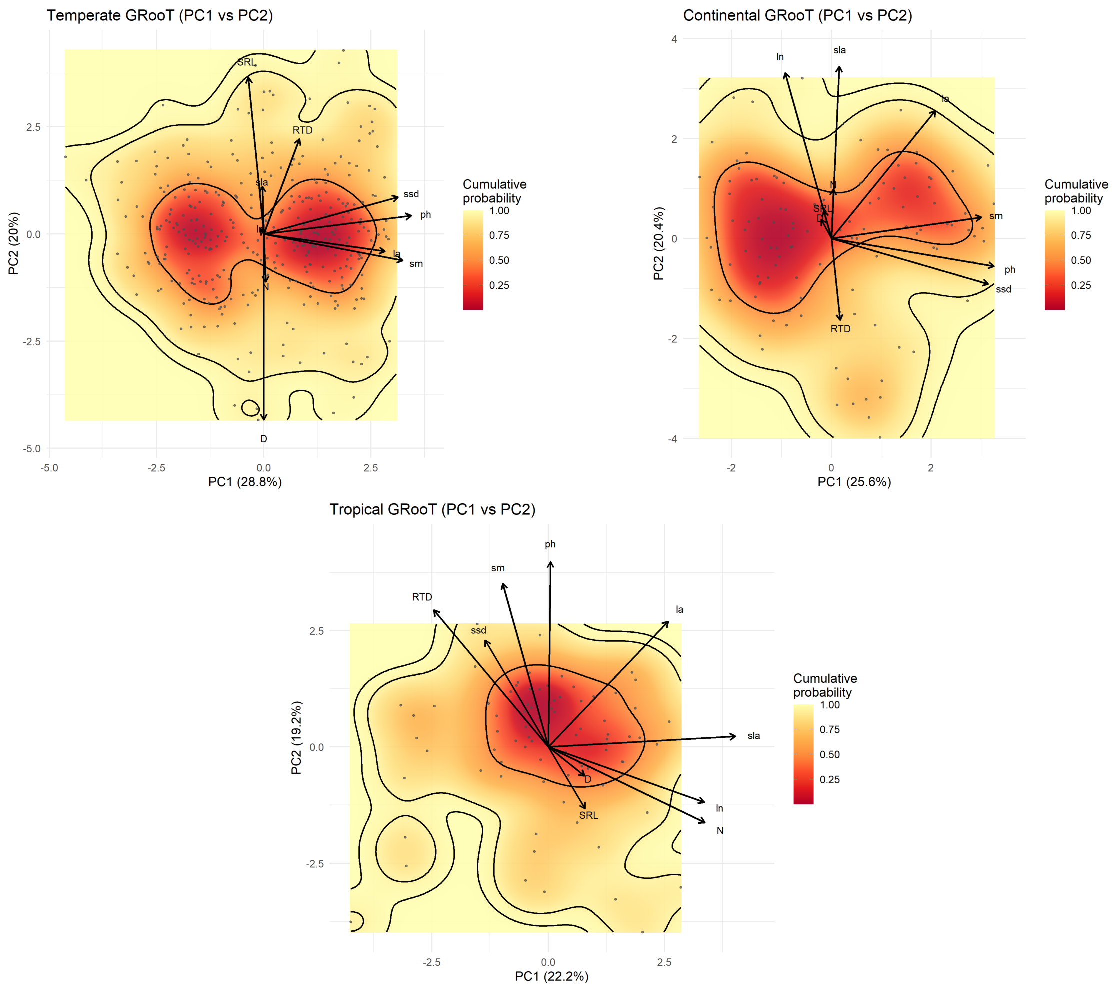

Figure 12: Functional space (PC1–PC2) for continental (A), temperate (B) and tropical (C) biome according to the GRooT biome separation.

Looking at the loadings of the PCAs in Fig. 13 a similar trend is detected. The temperate and continental biome seem to have the root traits simply shifted up to the PC2 component. An exception is the leaf area (la) trait of the temperate biome, which shifted to PC1. In comparison, the tropical biome depicts a completely different pattern, with nitrogen concentration being relevant on the PC1 direction and most aboveground traits shifted to PC2.

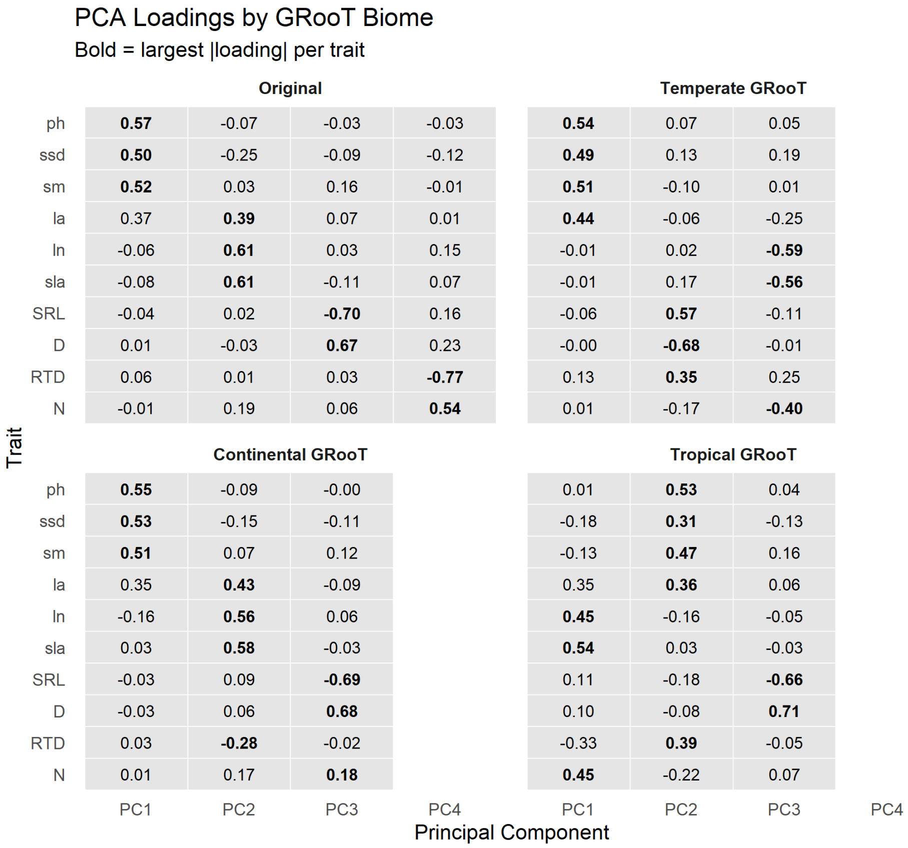

Figure 13: Loading comparison per biome

The second biome separation into the nine biomes, temperate grassland desert, temperate rain forest, temperate seasonal forest, tropical rain forest, tropical seasonal forest savanna, subtropical desert, boreal forest, woodland shrubland and tundra, is also analysed for comparison. PCA visualizations of these detailed biomes can be found in Appendix C. The loadings are listed in Fig. 14. The tundra biome only contained 14 data points, inhibiting a PCA analysis for the assigned species. The temperate grassland desert, temperate seasonal forest and woodland shrubland biomes are supporting the original claim of a separation into above- and fine-root trait spaces, with all aboveground traits on PC1 and PC2 and all fine-root traits on PC3 and PC4. The tropical seasonal forest savanna biome is the only tropical biome, that contained enough data for four principle components. For this biome the aboveground plane is the PC1-PC3 plane and the fine-root plane is the PC2-PC4 plane. A separation into two planes is nevertheless observed. The two rain forest biomes (temperate rain forest and tropical rain forest), as well as the subtropcial desert biome have mixed aboveground and root traits on both PC1 and PC2. The boreal forest shows similarities to the original PCA, and a separation into above- and fine-root planes could be expected after a data increase.

Visually, the PCA results (Appendix C) show a single trait space for the biomes temperate rain forest, subtropical desert and tropical rain forest, while all others, even though weakly, indicate a second trait space. These three biomes have their mostly constant environmental conditions in common. The single trait space could indicate that species of these biomes behave differently than concluded by Carmona et al. (2021). They might not depict two plant strategies, that can be appointed to herbaceous species and angiosperm trees, but instead have rather similar strategies.

Carmona et al. (2021) did not make statements regarding a biome-specific PCA analysis, but mentioned higher dissimilarities on the aboveground plane across all biomes. They support this claim with PERMANOVA test results (42.8% ± 18 aboveground) but mention that the variation aboveground is lower for biomes than it is for families. Our visual PCA analysis shows weaker visual separation on the aboveground plane, which needs to be further assessed.

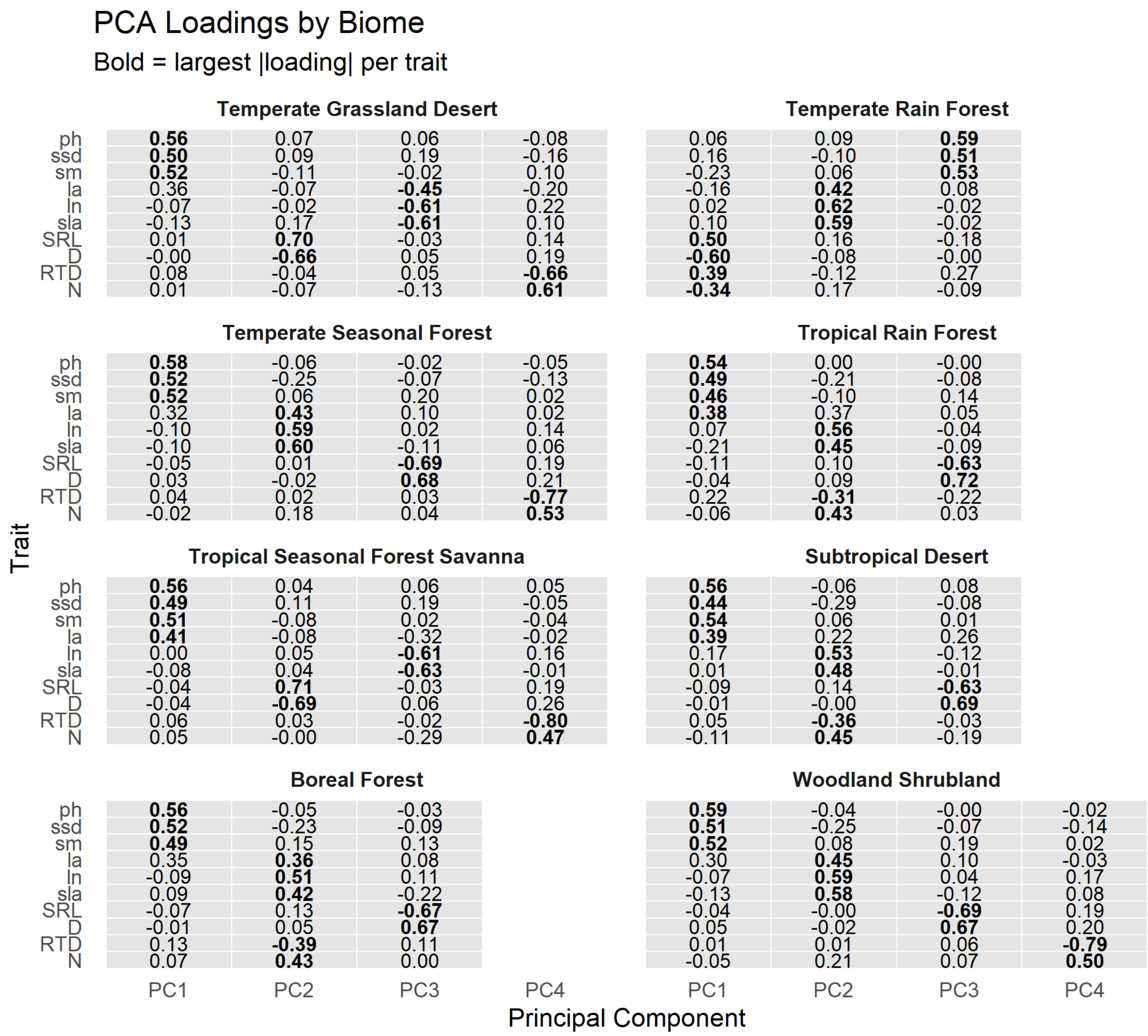

Figure 14: Loading comparison per biome

2.6 Conclusion biome universality

In conclusion, the conducted biome-specific PCA analysis shows that tropical biomes and biomes with more constant environmental conditions behave differently. Different loading patterns and only one functional trait space on the PC1-PC2 plane is observed for the tropical biome. While this observation is mainly associated with an insufficient data amount and needs to be verified with larger data sets per biome, it might still highlight an under-representation of these biomes in the publication of Carmona et al. (2021). It is highly likely that larger data sets for these biomes return to the observed patterns from Carmona et al. (2021), but we strongly recommend repeating this additional biome-specific analysis with larger data subsets, to rule out an incorrect assumption of regional universality.

3 Quantifying the robustness of PCA-derived trait spaces

3.1 Aim of the ordination analysis

The main objective of this study was to interpret biological structure in PCA ordinations derived from multivariate plant trait data. Spatial patterns in PCA space are commonly interpreted as reflecting functional differentiation, trait syndromes, or ecological gradients. Such interpretations, however, are only meaningful if the inferred structure is robust to methodological choices made during data processing and ordination.

PCA results are influenced by several analytical decisions, including the treatment of missing values (e.g. trait imputation), the selection and number of retained axes, and, critically, the application of axis rotation. These choices can affect not only axis orientation but also the apparent spatial configuration of observations in ordination space. Robustness checks are therefore required to ensure that observed spatial patterns reflect underlying trait relationships rather than artifacts of the analytical pipeline.

3.2 Procrustes test

To evaluate whether preprocessing steps or other analytical choices substantially alter the spatial configuration of species in the PCA space, a Procrustes analysis can be deployed. Rather than comparing individual axes or loadings, Procrustes analysis assesses whether species occupy similar relative positions across ordinations and whether the overall geometry of the trait space is preserved.

The method aligns one ordination to another through rotation, translation, and scaling, without modifying the internal structure of either configuration. Any remaining mismatch after alignment therefore reflects true differences in spatial configuration, not differences arising from axis orientation or scaling. This mismatch is summarized by a Procrustes similarity (or equivalently, a residual mismatch) and interpreted as an indicator of geometric stability.

High Procrustes similarity indicates that relative distances among species are conserved and that rotation or preprocessing does not fundamentally alter the trait space. Conversely, low similarity would suggest that spatial patterns observed in the rotated PCA may be partially artefactual and that biological interpretations should be treated with caution. In this way, Procrustes analysis functions as a formal robustness check linking methodological decisions to the validity of ecological inference.

3.3 Step 1: Testing whether varimax rotation alters PCA geometry

Why do we do this step? In this study, varimax rotation was applied to improve the interpretability of PCA axes by maximizing the variance of squared loadings, thereby concentrating each trait's contribution on a small number of components ("one-hot" loading structure). This facilitates biological interpretation by aligning components more closely with specific trait syndromes, while preserving the total variance explained by the retained axes (see Appendix D).

A common criticism of rotated ordinations is that this redistribution of variance may exaggerate distances between observations, introduce or amplify apparent clustering or alter the relative geometry of observations in multivariate space. As a result, there is a risk that biological interpretations based on rotated PCA axes could be driven by rotation-induced geometry rather than by genuine structure in the underlying trait data. Therefore, before extending or modifying the original analysis, it is necessary to verify that varimax rotation does not fundamentally change the spatial configuration of species in trait space.

How do we do it? This concern was explicitly raised during peer review of the original study by Carmona et al. (2021). In response, the authors validated the effect of varimax rotation using a Procrustes framework. They constructed three trait datasets (aboveground traits only, fine-root traits only, and a combined dataset), ran unrotated PCAs, and used Horn's parallel analysis to select dimensionality (2 axes for aboveground traits, 2 axes for fine-root traits, and 4 axes for the combined dataset). Varimax rotation was then applied to the selected axes to define interpretable "functional spaces".

To test whether rotation altered ordination geometry, the authors performed three Procrustes tests with 9,999 permutations, comparing: (i) the aboveground trait space with the C1–C2 plane of the combined space, (ii) the fine-root trait space with the C3–C4 plane of the combined space, and (iii) the aboveground and fine-root spaces directly. Procrustes analysis aligns two ordinations by rotation, translation, and scaling while preserving their internal structure; the resulting Procrustes correlation quantifies the similarity of the spatial configurations.

```r
library(paran)
library(ade4)

# Unrotated PCA (traits already log-transformed, centered, scaled)
pca_above <- prcomp(X_above)
pca_root  <- prcomp(X_root)
pca_all   <- prcomp(X_all)

# Dimensionality selection (Horn's parallel analysis)
paran(X_above)  # selects 2 axes
paran(X_root)   # selects 2 axes
paran(X_all)    # selects 4 axes

# Varimax rotation of selected axes
rot_above <- varimax(pca_above$rotation[, 1:2])
rot_root  <- varimax(pca_root$rotation[,  1:2])
rot_all   <- varimax(pca_all$rotation[,   1:4])

# Species scores in rotated spaces
scores_above <- as.matrix(X_above) %*% rot_above$loadings
scores_root  <- as.matrix(X_root)  %*% rot_root$loadings
scores_all   <- as.matrix(X_all)   %*% rot_all$loadings

# Procrustes tests (9999 permutations)
proc_above_vs_all12 <- procuste.rtest(scores_above, scores_all[, 1:2],
   nrepet = 9999)
proc_root_vs_all34  <- procuste.rtest(scores_root,  scores_all[, 3:4],
   nrepet = 9999)
proc_above_vs_root  <- procuste.rtest(scores_above, scores_root,
   nrepet = 9999)
```

Listing 5: Core analysis steps by Carmone et al. used to validate varimax rotation (schematic).

Table 1: Procrustes tests validating that varimax rotation does not induce artificial ordination structure (Carmona et al. 2021).

Comparison	Procrustes correlation	P
Aboveground space vs. combined C1–C2	0.988	0.0001
Fine-root space vs. combined C3–C4	0.982	0.0001
Aboveground space vs. fine-root space	0.178	0.0001

What do we learn from this step? The first two Procrustes tests show extremely high correspondence between the rotated aboveground and fine-root spaces and their respective planes in the combined four-dimensional space (Table 1). This demonstrates that varimax rotation does not alter the relative positions of species in trait space: the geometry of the ordination is already present in the unrotated PCA and is merely re-expressed in a more interpretable coordinate system.

The third Procrustes test shows low correspondence between the aboveground and fine-root spaces, supporting the biological interpretation that aboveground and belowground trait syndromes are only partially coupled. Importantly, this decoupling reflects genuine biological structure rather than a rotation artefact.

Although varimax rotation substantially changes trait loadings on individual axes , this does not affect ordination geometry. Rotation changes the basis within the retained PCA subspace but preserves all pairwise distances up to an orthogonal transformation. Procrustes analysis explicitly tests this property, making it an appropriate validation step. Together, these results establish that varimax rotation improves interpretability without introducing artificial spatial separation and provide a validated foundation for subsequent robustness analyses.

3.4 Step 2: Testing robustness to trait imputation

Why do we do this step? While robustness to varimax rotation has been demonstrated, the study introduces additional methodological steps that could potentially affect ordination geometry. The stability of these steps was qualitatively addressed before; here we quantify stability using Procrustes analysis. One such step is trait imputation, which increases data coverage but may also introduce uncertainty.

If imputation were to distort the geometry of trait space, species positions in the PCA could reflect imputation artefacts rather than empirical structure. It is therefore necessary to test whether imputation changes the underlying configuration of species in multivariate space.

How do we do it? To isolate the effect of trait imputation from artefacts caused by refitting the PCA, we compared two representations of the *same species* within a single, fixed PCA space. Specifically, the PCA fitted on the imputed combined dataset was used as a reference space (PC1–PC4), defining the axes, centering, scaling, and rotation.

Species with complete empirical trait data ($n = 301$) were then projected into this imputed reference space using only their observed trait values, applying the same standardization parameters and rotated loadings. This yields a second configuration for the same set of species, expressed in the same coordinate system but without relying on imputed trait values.

The projected non-imputed configuration was compared to the corresponding species configuration from the imputed PCA using Procrustes analysis (`ade4::procuste.rtest`, 9,999 permutations). Because both configurations are defined in the same PCA space, any remaining mismatch reflects differences introduced by imputation rather than differences in axis definition or PCA refitting.

```
library(ade4)

# Load reference PCA (imputed combined dataset) + not-imputed combined traits
ref    <- readRDS(file.path("Results", "PCA_imputed_combined_full.rds"))
notimp <- readRDS(file.path("Data", "not_imputed_combined.rds"))

ref_traits   <- as.data.frame(ref$traits)
```

```r
ref_scores   <- as.data.frame(ref$traitsUse)      # reference scores (imputed space)
ref_loadings <- as.matrix(ref$PCA$loadings)       # rotated loadings (traits x components)
ref_means    <- ref$means
ref_sds      <- ref$sds

# Align trait columns and order to reference
ref_trait_names <- colnames(ref_traits)
notimp_aligned  <- as.data.frame(notimp)[, ref_trait_names, drop = FALSE]

# IMPORTANT: standardize using REFERENCE means/sds (not not-imputed means/sds)
Z_not <- scale(notimp_aligned, center = ref_means, scale = ref_sds)

# Project into the same rotated PCA space
scores_not <- Z_not %*% ref_loadings
scores_not <- as.data.frame(scores_not)
rownames(scores_not) <- rownames(notimp_aligned)

# Match shared species IDs
common_ids <- intersect(rownames(ref_scores), rownames(scores_not))
X_ref <- ref_scores[common_ids, 1:4, drop = FALSE]
X_not <- scores_not[common_ids, 1:4, drop = FALSE]
X_not <- X_not[rownames(X_ref), , drop = FALSE]

# Procrustes test (permutation)
set.seed(1)
proc_res <- ade4::procuste.rtest(X_ref, X_not, nrepet = 9999)
print(proc_res)
```

Listing 6: Key code lines (Step 2): project non-imputed species into the imputed reference PCA space and run Procrustes.

What do we learn from this step? The Procrustes test revealed a highly significant correspondence between the projected non-imputed configuration and the imputed reference configuration ($R = 0.974$, $p = 0.0001$; 2). In the applied implementation, the reported statistic represents the residual mismatch between configurations (expressed here as a similarity measure; higher values indicate closer match), which was far smaller than expected under random permutation.

This result shows that imputation does not introduce a different trait-space geometry for the shared species. Instead, imputation primarily increases completeness and coverage, while empirically measured species retain consistent relative positions in the PCA space. This validates the imputed reference ordination as a stable basis for the subsequent robustness steps.

	Test configuration	Scope	n	Axes compared	Statistic
1	Rotation	Above vs. Below domains	–	C1–C2 / C3–C4	ProcCorr = 0.988 / 0.982
1	Rotation	Cross-domain	–	C1–C2 vs. C3–C4	ProcCorr = 0.178
2	Imputation	Global (projected)	301	PC1–PC4	ProcCorr = 0.979
3	OOB (oob_1x)	Global	1218	PC1–PC4	Mean $m^2 = 0.058$
3	OOB (oob_2x)	Global	1218	PC1–PC4	Mean $m^2 = 0.058$
4	Biome: Temperate	Groot PCA	71	PC1–PC3	ProcCorr = 0.99
4	Biome: Tropical	Groot PCA	27	PC1–PC3	ProcCorr = 0.88
4	Biome: Continental	Groot PCA	40	PC1–PC3	ProcCorr = 0.98

Table 2: Summary of Procrustes analyses assessing robustness of PCA-derived trait spaces. For each test, the number of shared species (n), the PCA axes compared, the Procrustes statistic (correlation for configuration similarity or residual m^2 for OOB), and permutation-based p-values (9,999 permutations) are reported. High Procrustes correlation or low residual mismatch indicates strong geometric correspondence. Biome analyses are based on combined (groot) trait PCAs; axes compared reflect the minimum shared dimensionality between biome and global reference spaces.

3.5 Step 3: Testing stability under imputation uncertainty

Why do we do this step? Even if trait imputation does not systematically distort ordination geometry (Step 2), uncertainty in imputed values could still affect the stability of species positions in PCA space. In particular, stochastic variation introduced during imputation may propagate into the ordination and blur or shift spatial patterns.

To evaluate whether imputation uncertainty introduces structural changes in PCA geometry, rather than random noise, we explicitly tested ordination stability under different levels of imputation uncertainty.

How do we do it? Imputation uncertainty was quantified using an out-of-bag (OOB) framework from the random forest–based `missForest` imputation procedure. The OOB error reflects prediction uncertainty for missing values based on trees that did not use a given observation during model fitting. Two uncertainty regimes were considered: a baseline OOB scenario (`oob_1x`) and an inflated uncertainty scenario in which OOB variability was doubled (`oob_2x`).

For each OOB-imputed replicate dataset: (i) species scores were computed in the same PCA space (PC1–PC4), (ii) aligned to a fixed reference PCA configuration, and (iii) compared to the reference using Procrustes analysis.

For each comparison, we recorded the Procrustes residual m^2 as a measure of geometric mismatch between the OOB-imputed configuration and the fixed reference PCA space. The residual m^2 represents the remaining sum of squared distances between corresponding species after optimal superimposition (translation, rotation, and uniform scaling).

In contrast to earlier steps, we used an alternative Procrustes implementation that directly returns the residual m^2 and does not perform permutation testing. This choice was motivated by computational efficiency: because a large number of bootstrap replicates are compared against a single fixed reference configuration, repeated permutation tests would be computationally expensive and statistically redundant.

In this context, the goal is not hypothesis testing between two independent ordinations, but quantifying the magnitude of deviation introduced by imputation uncertainty. The residual m^2 therefore serves as a continuous stability metric: lower values indicate closer alignment to the reference configuration and greater positional stability of species.

For comparability with previous analyses based on Procrustes correlation, we transformed the residual m^2 into an R-equivalent similarity measure. Under the standard normalization of Procrustes analysis, this relationship can be expressed as

$$R = \sqrt{1 - m^2},$$

where $R \in [0, 1]$ increases with structural similarity and m^2 decreases with geometric mismatch.

```
library(ade4)

# Load reference PCA scores
ref_scores <- readRDS(file.path("Results", "PCA_reference_scores_PC1_4.
    rds"))

# Loop over OOB replicate sets
results <- list()

for (set in c("oob_1x", "oob_2x")) {

  oob_files <- list.files(
    file.path("Results", "OOB_imputation", set),
    pattern = "\\.rds$",
    full.names = TRUE
  )

  for (f in oob_files) {

    # Load OOB-imputed species scores
    oob_scores <- readRDS(f)

    # Align species
    common_ids <- intersect(rownames(ref_scores), rownames(oob_scores))
    X_ref <- ref_scores[common_ids, 1:4, drop = FALSE]
    X_oob <- oob_scores[common_ids, 1:4, drop = FALSE]
    X_oob <- X_oob[rownames(X_ref), , drop = FALSE]

    # Procrustes comparison
    proc <- procuste(X_ref, X_oob)

    results[[length(results) + 1]] <- data.frame(
      set = set,
      obs = proc$ss
    )
  }
}

results_df <- do.call(rbind, results)
summary(results_df)
```

Listing 7: Key code lines (Step 3): OOB-based Procrustes stability analysis.

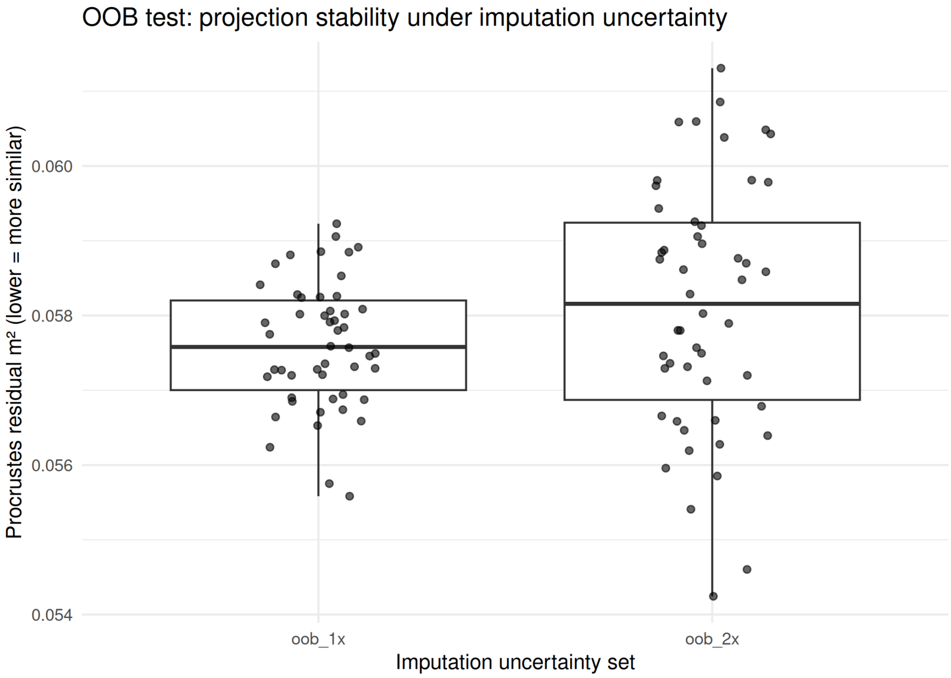

Figure 15: Distribution of Procrustes residuals (m^2) for OOB-based imputation replicates under baseline (`oob_1x`) and inflated (`oob_2x`) uncertainty regimes. Lower values indicate higher geometric similarity to the reference PCA space.

What do we learn from this step? Procrustes residuals were small and nearly identical across the two uncertainty regimes (Fig. 15; Table 2). Even when imputation uncertainty was intentionally increased, the relative positions of species in PCA space remained highly stable.

This demonstrates that imputation uncertainty introduces random noise but does not induce systematic or structural changes in ordination geometry. Consequently, the global gradients and spatial configuration of species in the PCA-derived trait space are robust to realistic and exaggerated levels of imputation uncertainty.

3.6 Step 4: Testing biome-specific ordinations against the global trait space

Why do we do this step? We asked whether biome-specific trait spaces follow the same global gradients, or whether some biomes—especially underrepresented ones—exhibit a distinct internal structure. Because PCA is refitted within each biome, biome-specific ordinations can differ from the global one for two separable reasons: (i) *dimensional collapse*, where trait variation within a biome concentrates into fewer independent gradients (a simpler landscape), and (ii) *structural reorganization*, where relative species positions change such that the biome no longer aligns with the global spectrum.

Dimensional collapse alone does not imply a new functional landscape: gradients may compress, and above- vs. belowground separation may weaken, while species relationships can still remain consistent with the global configuration. To distinguish compression from reorganization, we used Procrustes analysis, which compares the *relative positions of species (scores)* in trait space, not loadings or axis meaning.

Under this logic, *low* Procrustes similarity would indicate biome-specific ecological filtering that reshapes species relationships (a different landscape), whereas *moderate to high* similarity would indicate that biomes occupy compressed subsets of the global spectrum rather than forming qualitatively new configurations.

How do we do it? We used the global PCA fitted on the imputed combined dataset as a fixed reference trait space. For each biome (temperate, tropical, continental), we fitted a biome-specific PCA using the same preprocessing pipeline. Kathi's results showed that some biomes retain fewer effective dimensions; therefore, we compared each biome-specific ordination to the global reference using only the number of axes supported by the biome PCA.

Concretely, for each biome we: (i) determined k_b, the number of retained biome-specific components, (ii) extracted the biome species' *scores* on the first k_b biome axes, (iii) extracted the same species' scores on the first k_b axes of the global reference PCA, and (iv) quantified geometric correspondence using Procrustes correlation (R; 9,999 permutations).

This ensures that both configurations contain the same species and the same dimensionality, and that any mismatch reflects differences in species geometry rather than differences caused by comparing spaces of unequal dimension.

```r
library(ade4)

global_scores <- readRDS(file.path("Results", "PCA_imputed_combined_scores.rds"))
biome_meta    <- readRDS(file.path("Data", "species_biome_metadata.rds"))
trait_matrix  <- readRDS(file.path("Data", "trait_matrix_preprocessed.rds"))

biomes <- c("temperate", "tropical", "continental")
res <- list()

# --- helper: choose number of retained axes for biome PCA ---
# Replace this rule with your actual criterion (e.g., eigenvalue > 1, broken-stick, variance threshold)
choose_k <- function(pca_obj, min_k = 2, max_k = 4) {
  # example: retain up to 4 axes, but at least 2
  k <- min(max_k, ncol(pca_obj$x))
  k <- max(min_k, k)
  return(k)
}

for (b in biomes) {

  sp_b <- biome_meta$species[biome_meta$biome == b]
  sp_b <- intersect(sp_b, rownames(global_scores))
  sp_b <- intersect(sp_b, rownames(trait_matrix))

  X_b  <- trait_matrix[sp_b, , drop = FALSE]
  pca_b <- prcomp(X_b, center = TRUE, scale. = TRUE)

  k_b <- choose_k(pca_b)  # biome-specific retained dimensionality

  # Compare SCORES for the SAME species, in the SAME dimensionality
  B_k <- pca_b$x[, 1:k_b, drop = FALSE]
  G_k <- global_scores[sp_b, 1:k_b, drop = FALSE]

  proc <- procuste.rtest(G_k, B_k, nrepet = 9999)

  res[[b]] <- data.frame(
    biome   = b,
    n       = length(sp_b),
    k       = k_b,
```

```
    ProcCorr = proc$obs,
    p_value  = proc$pvalue
  )
}

biome_results <- do.call(rbind, res)
print(biome_results)
```

Listing 8: Key code lines (Step 4): biome PCA vs global reference with biome-specific dimensionality k_b.

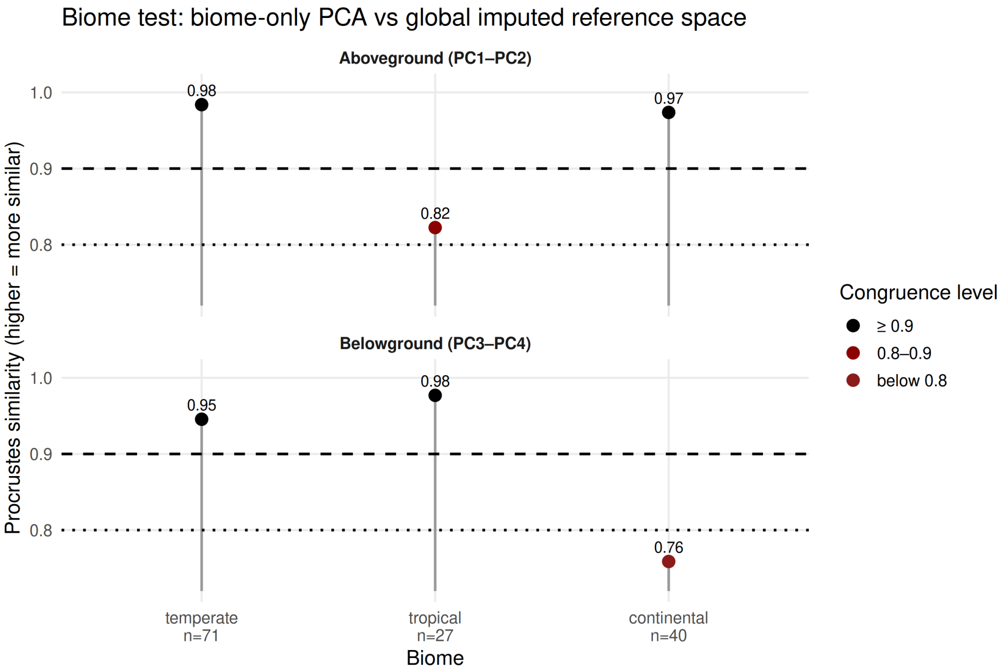

Figure 16: Procrustes similarity between biome-specific ordinations and the global imputed reference space. Higher R indicates greater correspondence of species score configurations.

What do we learn from this step? Biome-specific PCAs retained fewer effective dimensions in some cases, indicating that trait variation collapses into fewer independent gradients and the landscape becomes simpler. However, Procrustes comparisons showed moderate to high similarity to the global reference configuration (Fig. 16), meaning that dimensional collapse does not translate into a fundamentally different trait-space geometry.

Ecologically, this suggests that environmental filtering can *compress* the global trait landscape within biomes—reducing dimensionality and weakening above- vs. belowground separation—without *reorganizing* species relationships into a new functional landscape. In other words, biomes tend to represent simplified subsets of the global spectrum rather than qualitatively distinct configurations.

3.7 Overall Conclusion

Across all steps, Procrustes analyses consistently show that the PCA-derived trait space is robust to varimax rotation, trait imputation, imputation uncertainty, and data subsetting. Where deviations occur, they are domain-specific and biologically interpretable rather than artefacts of the analytical pipeline.

A Appendix: PCA glossary

To support interpretation of our multivariate analyses, key PCA-related concepts are explained in three complementary ways:

(1) Technical definition (general statistical meaning)
(2) Landscape metaphor (map and cities analogy)
(3) Specific interpretation in the context of our study

A.1 Multivariate Space ("Trait Space")

Technical. A multivariate space is a geometric space in which each dimension corresponds to one variable. Any observation measured on p variables can be represented as a point in a p-dimensional Euclidean space.

Landscape metaphor. The full landscape itself. Each data point is a city located somewhere within that landscape. The landscape may have many dimensions that cannot be directly visualized.

In our study. The multivariate space consists of ten standardized plant functional traits. Each species is represented as a point in this 10-dimensional space. This complete space contains the full ecological differentiation among species, even though we can only visualize reduced projections of it.

A.2 Imputation

Technical. Imputation is the statistical estimation of missing values using predictive models based on observed data patterns.

Landscape metaphor. Estimating the position of cities whose coordinates are missing by using information from nearby cities and the overall landscape structure.

In our study. Missing trait values of species were imputed using random forest models.

A.3 Principal Component Analysis (PCA)

Technical. PCA is a linear transformation that identifies orthogonal directions (principal components) maximizing variance in descending order. It provides a new coordinate system in which dimensions are mostly uncorrelated and ordered by explained variance.

Landscape metaphor. A cartographic method that chooses the most informative viewing angles from which to project the landscape onto map that is "as flat as possible".

In our study. PCA identifies the dominant gradients structuring global plant functional variation. Its goal is not merely dimensionality reduction, but the detection of major biological gradients organizing aboveground and fine-root traits across species.

A.4 Principal Components (PC1, PC2, . . .)

Technical. Principal components are orthogonal axes representing directions of maximal variance in the data. PC1 captures the greatest possible variance; each subsequent PC captures maximal remaining variance under orthogonality constraints.

Landscape metaphor. Newly defined cardinal directions (e.g., North–South, East–West) placed onto the landscape such that they span the largest variation across cities.

In our study. The first four components represent major axes of plant functional differentiation.

A.5 Ordination

Technical. An ordination is the configuration of observations in reduced-dimensional space resulting from a multivariate method such as PCA.

Landscape metaphor. The finished map showing all cities positioned relative to each other.

In our study. The ordination corresponds to the plotted configuration of species in planes such as C1–C2 or C3–C4. PCA is the method; the ordination is the resulting geometric representation.

A.6 Scores

Technical. Scores are the coordinates of observations projected onto the principal component axes.

Landscape metaphor. The exact x–y coordinates of each city on the map.

In our study. Species scores determine their positions in functional space and form the basis for distance calculations, comparisons and further analysis.

A.7 Loadings

Technical. Loadings are the coefficients of the linear combinations that define each principal component. They describe how the new axes are oriented relative to the original variable space. In other words, they specify how strongly each variable contributes to a component.

Landscape metaphor. Loadings describe what changes when moving in a given direction across the landscape. If walking north means that temperature decreases and precipitation increases, those relationships define what "north" represents. They define the meaning of a direction, not the positions of cities.

In our study. Principal components are statistical directions of maximal variance. Loadings allow us to interpret these directions biologically. For example, when traits related to plant size load strongly on one axis, that axis can be interpreted as a size gradient. Importantly, loadings describe axis meaning; they do not determine similarity among species. Similarity is determined by distances between species scores, not by loadings themselves.

A.8 Distance Between Points

Technical. Distance in ordination space is a geometric quantity (typically Euclidean distance) between observations based on their coordinates along retained components. It reflects multivariate dissimilarity.

Landscape metaphor. Distance corresponds to straight-line geographic distance between cities on the map. It is purely geometric. The fact that moving north becomes colder is a property of direction (defined by loadings); the distance between two cities measures how far apart they are in the mapped landscape, not whether they differ along a single environmental feature.

In our study. Distances between species in ordination space quantify overall functional dissimilarity across multiple traits simultaneously. Two species close together have similar multivariate trait combinations; species far apart differ across one or several dominant gradients. Distance reflects integrated ecological difference, not similarity along a single component.

A.9 Explained Variance

Technical. Explained variance is the proportion of total variance in the dataset accounted for by a principal component. It quantifies how much of the total multivariate structure is captured by that axis.

Landscape metaphor. It indicates how much of the full landscape structure becomes visible when projecting it from a particular viewing angle. Some directions reveal large terrain differences; others reveal only minor variation.

In our study. The proportion of variance explained by the first four components indicates how much of the global functional differentiation among species can be summarized by these axes. High explained variance for C1–C4 means that major global trait gradients are captured in reduced dimensions, allowing meaningful ecological interpretation without retaining all ten original variables.

A.10 Rotation (Varimax)

Technical. Varimax rotation is an orthogonal transformation applied to retained principal components. It redistributes variance among them to maximize loading sparsity — that is, it increases the tendency for each variable to load strongly on one component and weakly on others. This produces a more "one-hot" structure of loadings while preserving orthogonality and Euclidean distances among observations.

Landscape metaphor. Initially, the map may be oriented such that each compass direction captures a mixture of terrain features — moving northeast might mean it becomes slightly colder, slightly wetter, and slightly steeper all at once.

Rotation turns the map so that each direction aligns more cleanly with a dominant landscape feature. After rotation, moving north might correspond primarily to elevation, while moving east corresponds primarily to humidity. The cities do not move and the landscape does not change. Only the coordinate system becomes more aligned with interpretable gradients.

In our study. Before rotation, components may represent statistical mixtures of aboveground and belowground traits. Varimax rotation encourages a structure in which each trait loads strongly on one dominant axis and weakly on others. This makes axes correspond more clearly to coherent biological syndromes (e.g., predominantly aboveground vs predominantly fine-root gradients). Rotation therefore enhances interpretability, not ecological structure. It clarifies how existing variation is organized, without altering species similarity or functional distances.

A.11 Procrustes Analysis

Technical. Procrustes analysis quantifies similarity between two point configurations by optimally superimposing them through rotation, translation, and scaling. The resulting statistic measures how similar their geometric structures are.

Landscape metaphor. Two maps of the same region are placed on top of each other. After allowing rotation and resizing, the question is whether the cities occupy comparable relative positions.

In our study. Procrustes analysis evaluates whether two ordination spaces share the same underlying geometric structure. It compares configurations of species positions, not axes, not loadings, and not explained variance. A high Procrustes correlation indicates that relative species relationships are preserved between spaces. A low correlation indicates structural reorganization of the landscape. The method cannot identify which specific traits drive differences; it only evaluates similarity of overall configuration.

B Appendix: missForest imputation

"The method iteratively imputes each variable with missing values by fitting a random forest on the observed part of that variable and the current imputations of all other variables. After each iteration, the difference between the current and previous imputed matrices is computed separately for numeric and factor columns. The stopping rule is met once both differences have increased at least once (or only the present type increases if there is only one type). In that case, the previous imputation (before the increase) is returned. Otherwise, the process stops at maxiter." (Stekhoven, 2011)

C Appendix: PCA visualizations for biome tests

PCA Loadings by Dataset Size

Bold = largest |loading| per trait

Full

Trait	PC1	PC2	PC3	PC4
ph	**0.57**	-0.07	-0.03	-0.03
ssd	**0.50**	-0.25	-0.09	-0.12
sm	**0.52**	0.03	0.16	-0.01
la	0.37	**0.39**	0.07	0.01
ln	-0.06	**0.61**	0.03	0.15
sla	-0.08	**0.61**	-0.11	0.07
SRL	-0.04	0.02	**-0.70**	0.16
D	0.01	-0.03	**0.67**	0.23
RTD	0.06	0.01	0.03	**-0.77**
N	-0.01	0.19	0.06	**0.54**

Half

Trait	PC1	PC2	PC3	PC4
ph	**0.57**	-0.06	-0.04	-0.06
ssd	**0.51**	-0.26	-0.09	-0.11
sm	**0.52**	0.04	0.15	-0.02
la	0.35	**0.42**	0.06	-0.04
ln	-0.07	**0.59**	0.04	0.18
sla	-0.09	**0.62**	-0.08	0.07
SRL	-0.04	-0.01	**-0.70**	0.14
D	-0.00	-0.02	**0.68**	0.24
RTD	0.08	-0.01	0.00	**-0.72**
N	-0.01	0.16	0.05	**0.59**

Quarter

Trait	PC1	PC2	PC3	PC4
ph	**0.57**	-0.08	-0.05	
ssd	**0.49**	-0.27	-0.02	
sm	**0.52**	-0.02	0.18	
la	**0.40**	0.33	-0.01	
ln	-0.02	**0.55**	-0.01	
sla	-0.00	**0.50**	-0.23	
SRL	-0.05	0.09	**-0.66**	
D	0.01	0.10	**0.68**	
RTD	0.06	**-0.36**	-0.09	
N	-0.04	**0.33**	0.02	

Tenth

Trait	PC1	PC2	PC3	PC4
ph	**0.56**	-0.10	0.01	
ssd	**0.45**	-0.32	-0.02	
sm	**0.53**	-0.04	0.14	
la	**0.42**	0.26	0.01	
ln	-0.02	**0.52**	-0.14	
sla	-0.10	**0.45**	-0.24	
SRL	-0.08	0.09	**-0.63**	
D	-0.00	0.08	**0.69**	
RTD	0.07	**-0.43**	-0.16	
N	0.04	**0.38**	0.07	

Principal Component

Figure 17: PCA Loadings for different data amounts of the imputed dataset with (full) 1218 species

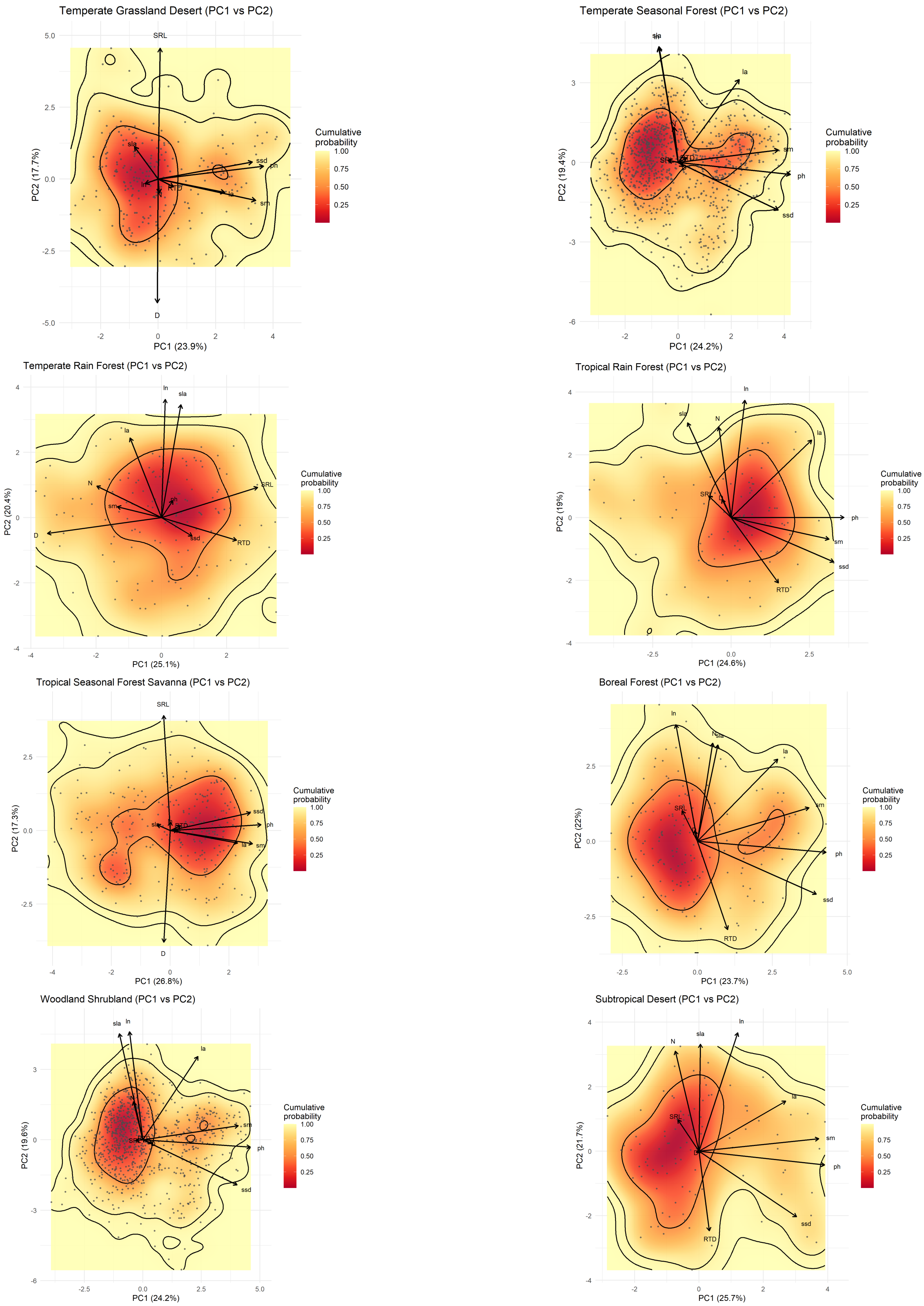

Figure 18: Functional space (PC1–PC2) for the data separation into 9 biomes. The biome Tundra PCA is not depicted as it contained too little data to conduct a PCA analysis.

D Appendix: Varimax Rotation

Varimax rotation is an orthogonal post-processing step applied to a retained set of principal components. It does *not* change the underlying geometry of observations (species scores) but reorients the axes to improve interpretability.

In PCA, the initial components maximize explained variance. However, the resulting loading structure may be difficult to interpret, because many variables can load moderately on multiple axes. Varimax rotation redistributes variance across retained components to achieve a simpler structure: each variable loads strongly on one component and weakly on others. This produces a more “one-hot” loading pattern while preserving orthogonality.

Importantly, because varimax is an orthogonal rotation,

$$\mathbf{T}^\top \mathbf{T} = \mathbf{I},$$

distances among observations remain unchanged. The rotation alters the coordinate system, not the relative configuration of points.

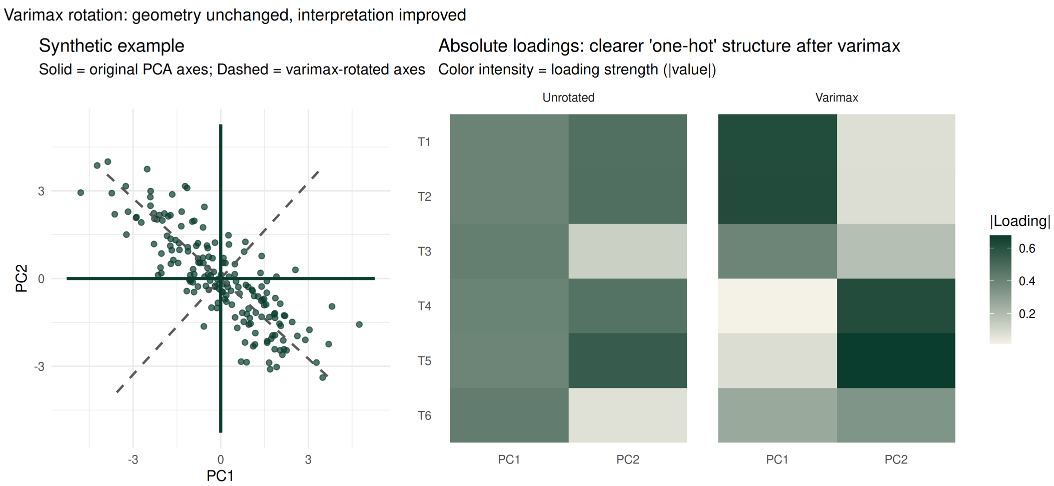

Figure 19: Synthetic illustration of varimax rotation. Left: identical point cloud with original (solid) and rotated (dashed) axes. Right: absolute loadings showing clearer, more concentrated structure after varimax.

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E Statement on the Use of AI

ChatGPT-5.1 and Claude Opus 4.5 were used for coding assistance and wording refinement.